HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · HOUSING FINANCE · CREDIT OPERATIONS

GENAI DOCUMENT INTELLIGENCE · HOME-LOAN UNDERWRITING

Loan File Intelligence

A GenAI assistant sorts every scanned page, extracts every field, cross-checks the documents and drafts the credit note, so underwriters spend their time on judgement, not reading.

~~120~~ → **38** underwriter minutes per file, indicative

11 types
Document types read automatically
KYC, income, banking, property

38 fields
Fields extracted per loan file
each with a confidence score

10 checks
Cross-document checks per file
name, income, property, LTV, FOIR

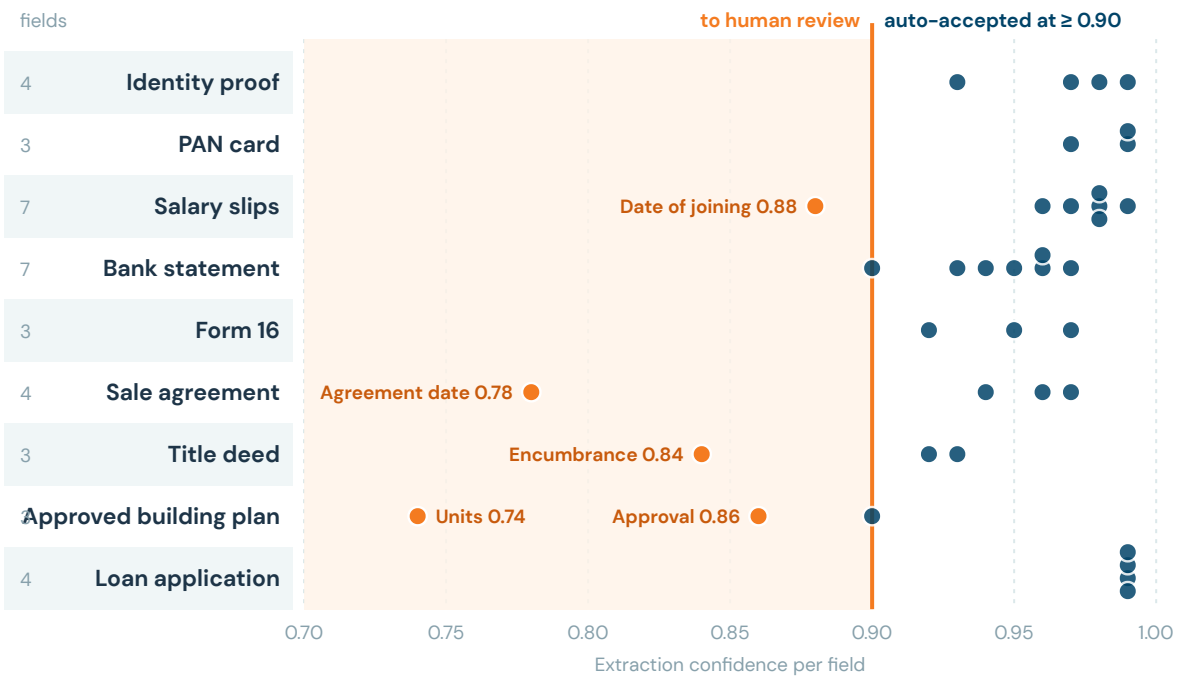
70 %
Underwriter reading time saved
indicative, from the pilot design

Rings: one segment per document type; the sample file's 38 fields, with the 5 below the 0.90 review threshold in orange; one segment per check; reading time saved.

01 Read the file: scans in, structured fields out

Synthetic sample loan file · identity numbers masked

11 scanned pages, random order
4 categories: KYC, income, banking, property
9 documents, each with a JSON schema
38 fields, each with a confidence score
10 cross-document checks, explained
1 draft credit note for the underwriter

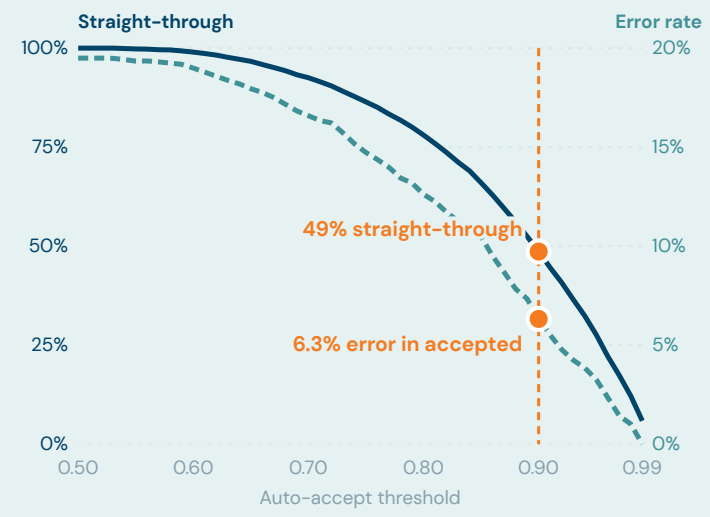


Each dot is one extracted field. Five fall below the 0.90 auto-accept line and go to a human reviewer; the other 33 pass straight through.

THE HUMAN-REVIEW DIAL

Higher threshold, fewer errors, more review

Fields below the threshold go to a reviewer. At 0.90, 49% of fields pass straight through with 6.3% error among those accepted.



Threshold curve from a simulated validation set of 4,000 fields; illustrative.

02 Ten checks, six files, every flag explained

Check results by scenario · synthetic loan file

CHECK · SOURCES	Clean	Name variant	Income mismatch	Property mismatch	High LTV	EMI bounce
1. Applicant name across documents ID · PAN · slip · bank · agreement	✓	! review	✓	✓	✓	✓
2. Date of birth: ID vs PAN ID · PAN	✓	✓	✓	✓	✓	✓
3. Employer consistency Slips · Form 16 · bank narration	✓	✓	✓	✓	✓	✓
4. Net salary vs bank credits Slips · bank statement	✓	✓	✕ fail	✓	✓	✓
5. Income stability (3 months) Salary slips	✓	✓	✓	✓	✓	✓
6. EMI or cheque returns Bank statement	✓	✓	✓	✓	✓	✕ fail
7. Property identity across documents Agreement · title · plan	✓	✓	✓	✕ fail	✓	✓
8. Buyer on agreement is the applicant Agreement · application	✓	✓	✓	✓	✓	✓
9. Loan-to-value within cap Application · agreement	✓	✓	✓	✓	✕ fail	✓
10. FOIR within policy Application · slips · bank	✓	✓	✓	✓	✓	✓
File readiness score	100 · READY	95 · READY	90 · READY	90 · READY	90 · READY	90 · READY

Pass: consistent · Review: explainable difference for a human · Fail: resolve before sanction

INCOME MISMATCH · FAIL
June credit 24% short
Bank credit ₹1,12,000 against net pay ₹1,47,800. Ask for an explanation or a later statement.

PROPERTY MISMATCH · FAIL
Tower B vs Tower C
The agreement and title deed name different towers. Resolve before legal sign-off.

HIGH LTV · FAIL
82.4% against an 80% cap
₹70,00,000 requested on an ₹85,00,000 agreement value.

NAME VARIANT · REVIEW
Middle name missing
Bank name is a 67% token match: likely the same person; confirm with KYC.

GUARDRAILS
Answer only from the file, always cite, never decide
Questions get cited answers; credit decisions stay with the underwriter.

03 Four pain points, four fixes

Before → after

BEFORE
Many document types
Identity proofs, salary slips, Form 16, bank statements, agreements, deeds and plans, all as scans.

CLASSIFY ON ARRIVAL

AFTER
Every page sorted
Each page classified into a document type before anyone opens it.

BEFORE
Manual re-keying
Names, incomes, addresses and values typed by hand, with typos that surface later.

SCHEMA EXTRACTION

AFTER
Fields with confidence
Extracted to a JSON schema; low-confidence fields routed to a reviewer.

BEFORE
Late inconsistencies
A name variant, a salary that doesn't match the bank credit, a different tower on the title.

RULES ENGINE

AFTER
Ten checks, with sources
Pass, review or fail, each explained in plain language.

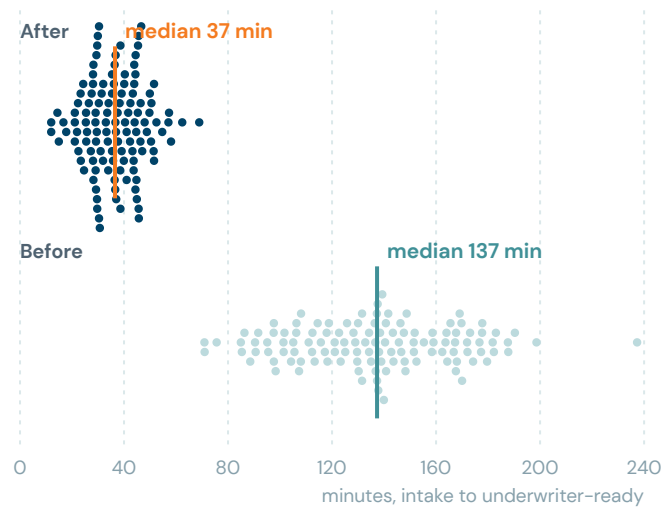
BEFORE
Judgement squeezed
Most underwriter time went into reading; credit notes written from scratch.

ASK · CITE · DRAFT

AFTER
Time back for decisions
Cited answers to questions and a drafted credit note to edit.

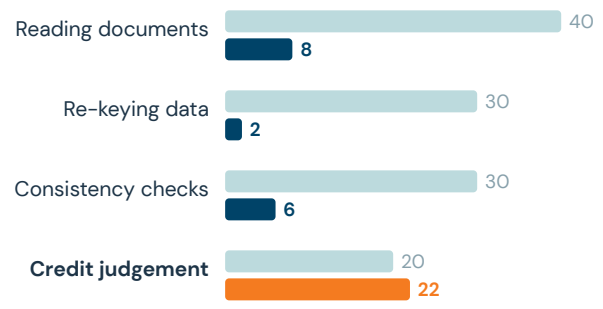
04 Review time per file

Each dot is one loan file · indicative



05 Where the minutes go

Underwriter minutes per file, before (light) and after · indicative



17% → 58% of underwriter time spent on credit judgement

06 How HMC ran it

2 weeks — Discover
Shadowed underwriters: document taxonomy and field list.

2 weeks — Design
JSON schema per document, ten checks, guardrails.

6 weeks — Build
Classify, extract, cross-check, ask-the-file, draft note.

Pilot — Adopt
Underwriters accept, edit or reject each flag.

∞ ongoing — Scale
New document types by schema, not code.

What any lender can borrow

01 — Extract to a schema, not to prose
Structured fields with confidence scores are what make checks, routing and audit possible.

02 — Cite everything
Underwriters trust an answer they can trace to a page; guardrails keep the assistant inside the file.

03 — Keep the human on the decision
Let GenAI read and reconcile; let thresholds route doubt to people, and leave credit judgement with them.

Built with **OCR** · **Large language model** · **Retrieval (embeddings)** · **Python rules engine** · **Human-in-the-loop review** | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

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